

Chapter 2

Business Processes – Sale Process

Learning objectives

- Master data and its role in ERP systems.
- Creating customer master data, material master data and pricing conditions.
- Sales and fulfillment cycle.
- Sales and Distribution in SAP S/4HANA (using Global Bike).

1. Sales and Service

Solution Map of SAP ERP

- The SAP ERP solution map

Functionality of SAP ERP

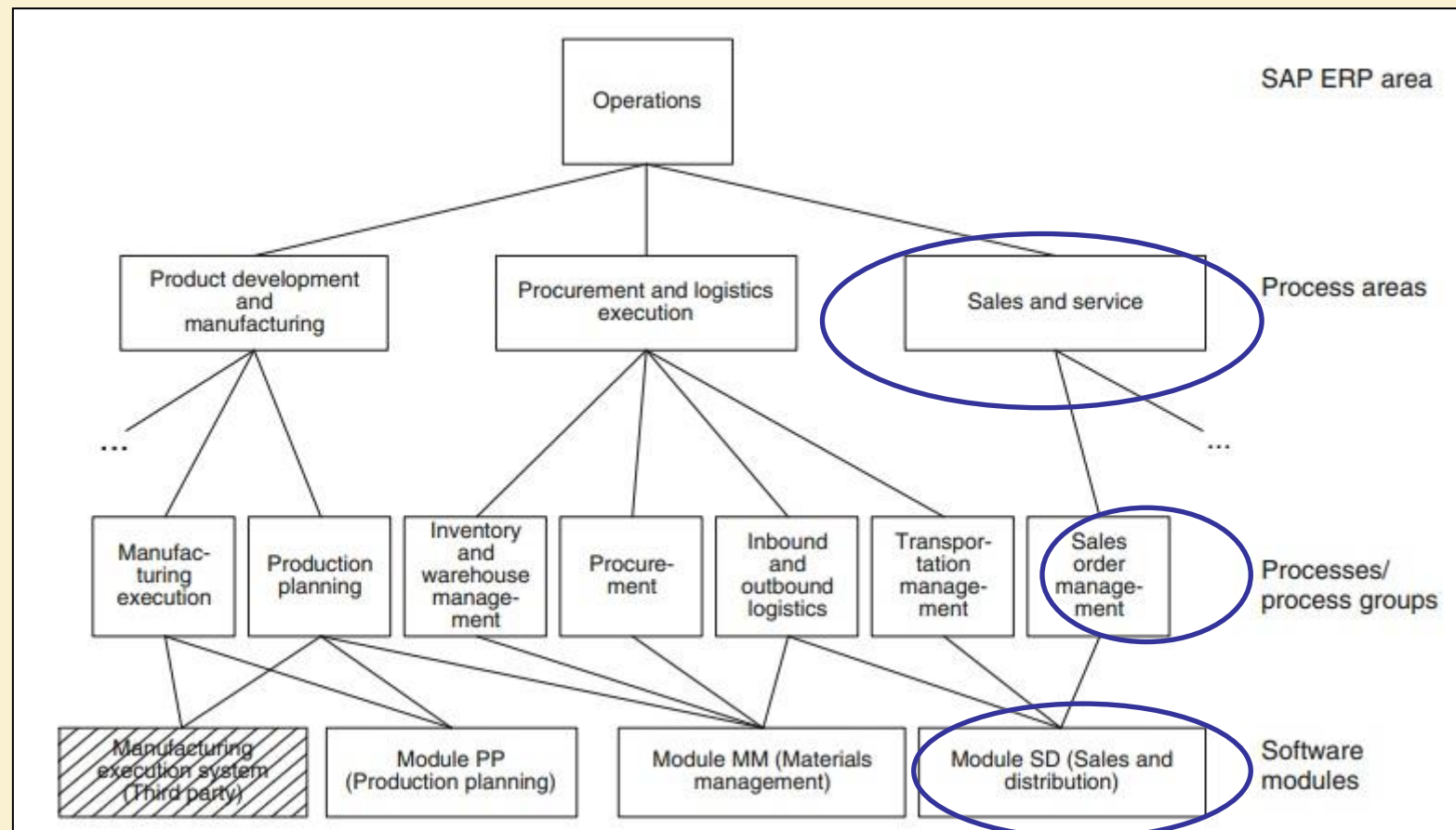
	End-user service delivery										Shared service delivery	SAP NetWeaver
Analytics	Financial analytics			Operations analytics			Workforce analytics					
Financials	Financial supply chain management		Treasury		Financial accounting		Management accounting		Corporate governance			
Human capital management	Talent management			Workforce process management			Workforce deployment					
Procurement and logistics execution	Procurement		Inventory and warehouse management		Inbound and outbound logistics			Transportation management				
Product development and manufacturing	Production planning		Manufacturing execution		Product development			Life-cycle data management				
Sales and service	Sales order management			Aftermarket sales and service			Professional-service delivery					
Corporate services	Real estate management	Enterprise asset management	Project and portfolio management	Travel management	Environment, health, and safety compliance management		Quality management	Global trade services				

Sales and Service

- Processes and functions for sales, distribution, and customer service belong to the sales and service part of the solution map. This part is broken down into:
 - ❖ Sales order management.
 - ❖ Aftermarket sales and service.
 - ❖ Professional service delivery.

Sales and Service (Cont.)

- Relationship between solution map and SAP ERP modules



Sales and Service – Sales Order Management

- Sales order management is the core and largest part of sales and service, supporting processes associated with sales and distribution tasks.
- The most important functions employed by sales order management are available in the Sales & Distribution module, focusing on quotation and order management:
 - ❖ Processing inquiries, quotations, pricing (taking into account conditions, surcharges, freights, taxes, etc.), credit card payment, credit checking, substitution products, compliance with minimum order quantities, availability checking, order scheduling, and incentive & commission management.

Sales and Service – Aftermarket Sales and Service

- Aftermarket sales and service support all aspects of service order processing within a service organization, from processing the initial inquiry to confirming the order and billing, all the way to product maintenance and long-term servicing.
- Among other things, aftermarket sales and service include installation and configuration management, contractual and warranty-based services, support for maintenance cycles, service requests, and service orders, as well as warranty and claims processing.

Sales and Service – Professional Service Delivery

- Professional service delivery is a subarea for managing consulting services.
- The capabilities provided are designed for selling, planning, delivering, and settling project-based services, including quotation and sales order processing.
- They support tasks related to consulting projects, such as project planning, project execution, and project accounting, as well as resource, time, and travel expense management.

2. Master Data and its role in ERP Systems

Master Data

- Master data are data that exist independent of specific orders (customer, production, purchase, transport orders, etc.).
- Master data constitutes the frame in which the planning and controlling of orders take place.
- Master data is the foundation of any business information system. Without reliable and robust master data, planning and controlling an enterprise is not possible.

Master Data (Cont.)

- Master data identifies and describes all the important business objects, for example, business partners, employees, articles, bills of materials, equipment, and accounts.
- Since all business activities, such as quotes, orders, postings, payment receipts, and transport orders, refer to the master data, these data are the basis of any coordination effort.

Master Data (Cont.)

- The most important *master data for production planning and control (MRP - Material Requirements Planning)* is data concerning:
 - ❖ Parts; Product structures; Operations; Routings.
 - ❖ Operating facilities or work centers; Manufacturing structures; Etc.
- Master data *for manufacturing resource planning (MRP II)*, in addition to those used in material requirements planning, include routings, operating facilities, factory calendars, shift models, tools, and employees, etc.

SAP S/4HANA - Data Types in ERP Systems

SAP S/4HANA Data Types in ERP Systems

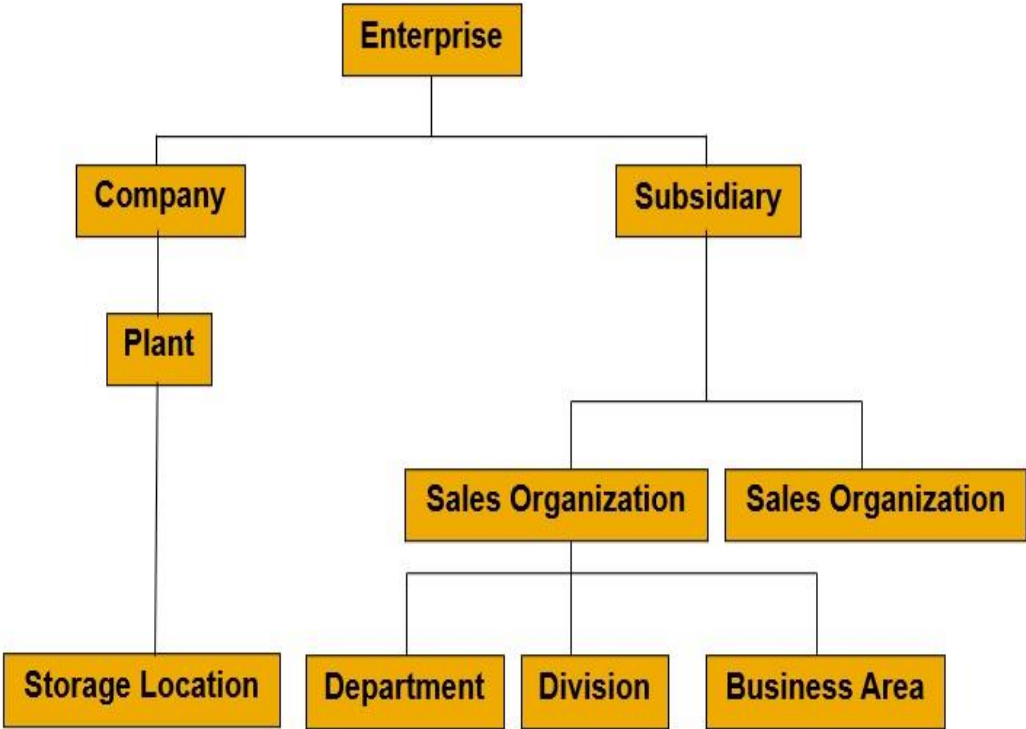
Organizational Data	Master Data	Transaction Data
Company Code	Person	Purchase Order
Plant	Material	Invoice
Storage Location	Customer	Quotation
Distribution Channel	Vendor	Sales Order
Purchasing Organization	Work Centre	Transportation Order
...	...	...

SAP S/4HANA - Data Types in ERP Systems

(Cont.)

SAP S/4HANA

Organizational Unit



SAP Terminology:

- ➡ Client
- ➡ Company Code
- ➡ Plant
- ➡ Sales Organization
- ➡ Division
Storage Location

SAP S/4HANA - Data Types in ERP Systems (Cont.)

SAP S/4HANA Master Data

- Stored for a long time and seldom changed
- Represent logically grouped data like:
 - Customer Master
 - Material Master
 - Vendor Master
 - General Ledger accounts

The screenshot displays the SAP S/4HANA interface for a Product Master Data record. The header shows the SAP logo, a 'Product' dropdown, and navigation icons. The main title is 'Deluxe Touring Bike (black)' with the material number 'DXTR1000'. Action buttons for 'Edit', 'Copy', and 'Share' are visible. Below the title, a blue cube icon represents the product. Key data fields include: Product Type: Finished Product (FERT), Base Unit of Measure: Each (EA), Revision Level, Product Category: Product, GTIN, and Product Group: Finished Bikes (BIKES). A navigation bar at the bottom lists tabs: General Information, Product Compliance, Components, Texts, Sales, Storage, and Warehouse Management. The 'General Information' tab is active, showing 'Basic Data' with fields for Division (Bicycles (BI)), Old Product Number, Batch Management Required (No), Marked for Deletion (No), Created By (Chris Reich), Created On (08/19/2021, 11:55:19), Last Changed By (Learn-900 Learn-900), and Last Changed On (09/06/2021, 16:18:36). A 'Show More' link is at the bottom right.

SAP S/4HANA - Data Types in ERP Systems (Cont.)

SAP S/4HANA Transaction Data

- Transaction data is the system record of business event.
- Depending on the business event, different master data and organizational data will be referenced
- For example, during a sales order business event, the following data is stored
 - Organizational level: client, company code, sales organization
 - Master data: customer, material, pricing (condition)
 - Situational data: date, time, person, amount

<  Create Inquiry: Overview

68

 More

Inquiry:

Net Value:

21,490.00

USD

Sold-To Party:

1003070

The Bike Zone 100 / 2144 N Orange Ave / Orlando FL 32804

Ship-To Party:

1003070

The Bike Zone 100 / 2144 N Orange Ave / Orlando FL 32804

Cust. Reference:

100

Cust. Ref. Date:

09/17/2021

Sales

Item Overview

Item detail

Ordering party

Procurement

Shipping

Reason for rejection

Valid From:

09/17/2021

Valid To:

10/17/2021

* Req. Deliv.Date:

D

10/17/2021

Expect.Ord.Val.:

0.00

USD



Group

All Items

	Item	Material	Req. Segment	Order Quantity	SU	Altltm
☐	10	DXTR1100			5 EA	
☐	20	PRTR1100			2 EA	
☐						

Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

SAP S/4HANA - Data Types in ERP Systems (Cont.)

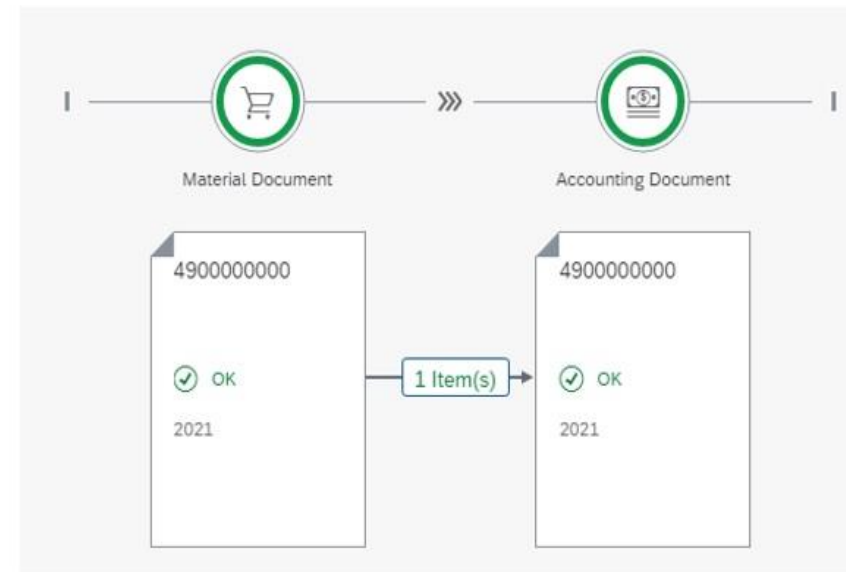
SAP S/4HANA Documents

- Transactions are data sets that are generated if a business transaction was executed.
- Is a record of the business transaction
- Includes all relevant predefined information from the master data and organizational entities
- Example:
 - Sales Document
 - Purchasing Document
 - Material Document
 - Accounting Document



Document Flow

- The document flow as well as the order status allow the setting of the status at any point in time
- SAP revises the status every time a change in a document takes place



3. Customer Master Data, Material Master Data, and Pricing Conditions.

Sales & Distribution Master Data

< SAP

Display Standard Order 5: Overview

6

📄

🔍

📊

Σ

More ▾

Standard Order: 5

Net Value: 20,092.50 USD

Sold-To Party: 1003063

The Bike Zone 000 / 2144 N Orange Ave / Orlando FL 32804

Ship-To Party: 1003063

The Bike Zone 000 / 2144 N Orange Ave / Orlando FL 32804

Cust. Reference: 000

Cust. Ref. Date: 09/14/2021

Customer Master

Sales

Item Overview

Item detail

Ordering party

Procurement

Shipping

Reason for rejection

Req. Deliv.Date: D 10/14/2021

Deliver.Plant:

Complete Dlv.: ☐

Total Weight: 57.170 G

Delivery Block: ▾

Volume: 0.000

Billing Block: ▾

Pricing Date: 09/14/2021

Pyt Terms: 0001 Pay immediately w/o deduction

Inco. Version:

Incoterms: FOB

Inco. Location1: Miami

Material Master

🔍

+

−

→

📊

📄

🔍

📄

📊

🔍

📄

📊

🔍

Group

🔍

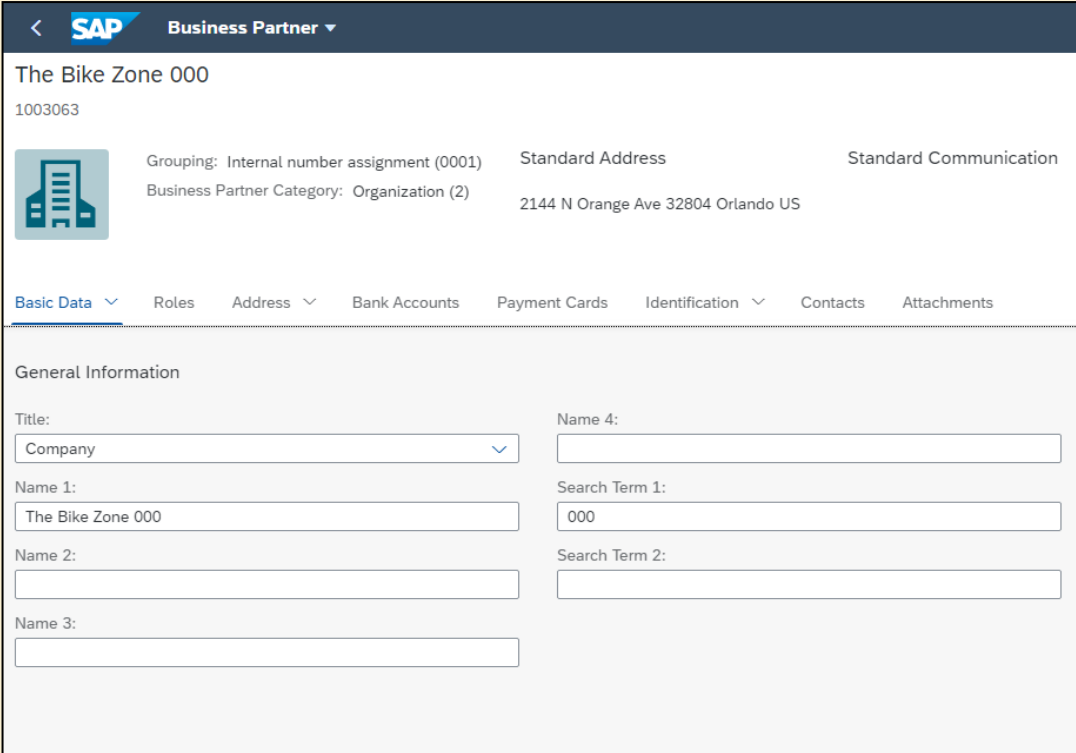
All Items

	Item	Material	Req. Segment	Order Quantity	Un	S	Item Description
☐	10	DXTR1000			5	EA	Deluxe Touring Bike (black)
☐	20	PRTR1000			2	EA	Professional Touring Bike (black)

Sales Condition

Customer Master Data

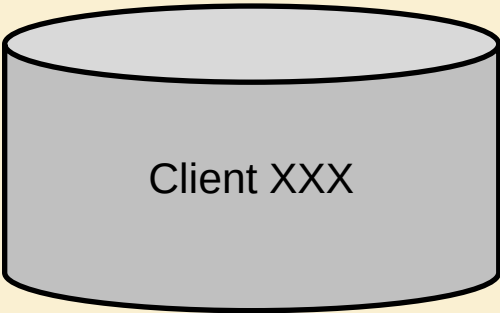
- Customer Master
 - ❖ Contains all of the information necessary for processing orders, deliveries, invoices and customer payment.
 - ❖ Every customer MUST have a master record.
 - But you can have one-time customers.
- Created by Sales Area
 - ❖ Sales Organization.
 - ❖ Distribution Channel.
 - ❖ Division.
- The customer master information is divided into 3 areas:
 - ❖ General Data.
 - ❖ Company Code Data.
 - ❖ Sales Area Data.



The screenshot displays the SAP Business Partner form for 'The Bike Zone 000' (ID 1003063). The header bar shows the SAP logo and 'Business Partner' with a dropdown arrow. Below the header, the customer name 'The Bike Zone 000' and ID '1003063' are shown. A building icon is next to the 'Grouping: Internal number assignment (0001)' and 'Business Partner Category: Organization (2)' fields. The 'Standard Address' field contains '2144 N Orange Ave 32804 Orlando US'. The 'Standard Communication' field is empty. A navigation bar below the header includes 'Basic Data' (selected), 'Roles', 'Address', 'Bank Accounts', 'Payment Cards', 'Identification', 'Contacts', and 'Attachments'. The 'General Information' section contains several input fields: 'Title' (dropdown menu with 'Company' selected), 'Name 1' (text field with 'The Bike Zone 000'), 'Name 2' (empty text field), 'Name 3' (empty text field), 'Name 4' (empty text field), 'Search Term 1' (text field with '000'), and 'Search Term 2' (empty text field).

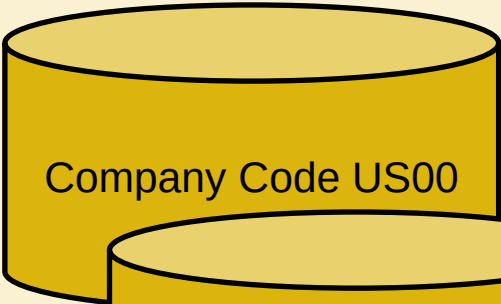
Customer Master

General Information relevant for the entire organization:

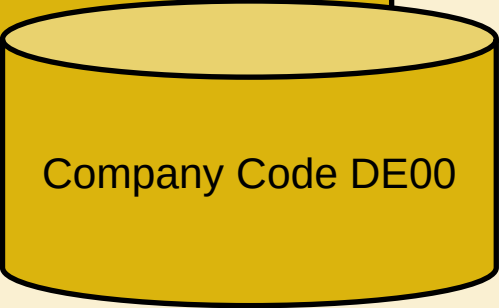


Name
Address
Communication

Company Code specific information:



Acc. Mgmt
Payment
Bank



Sales Area specific information:

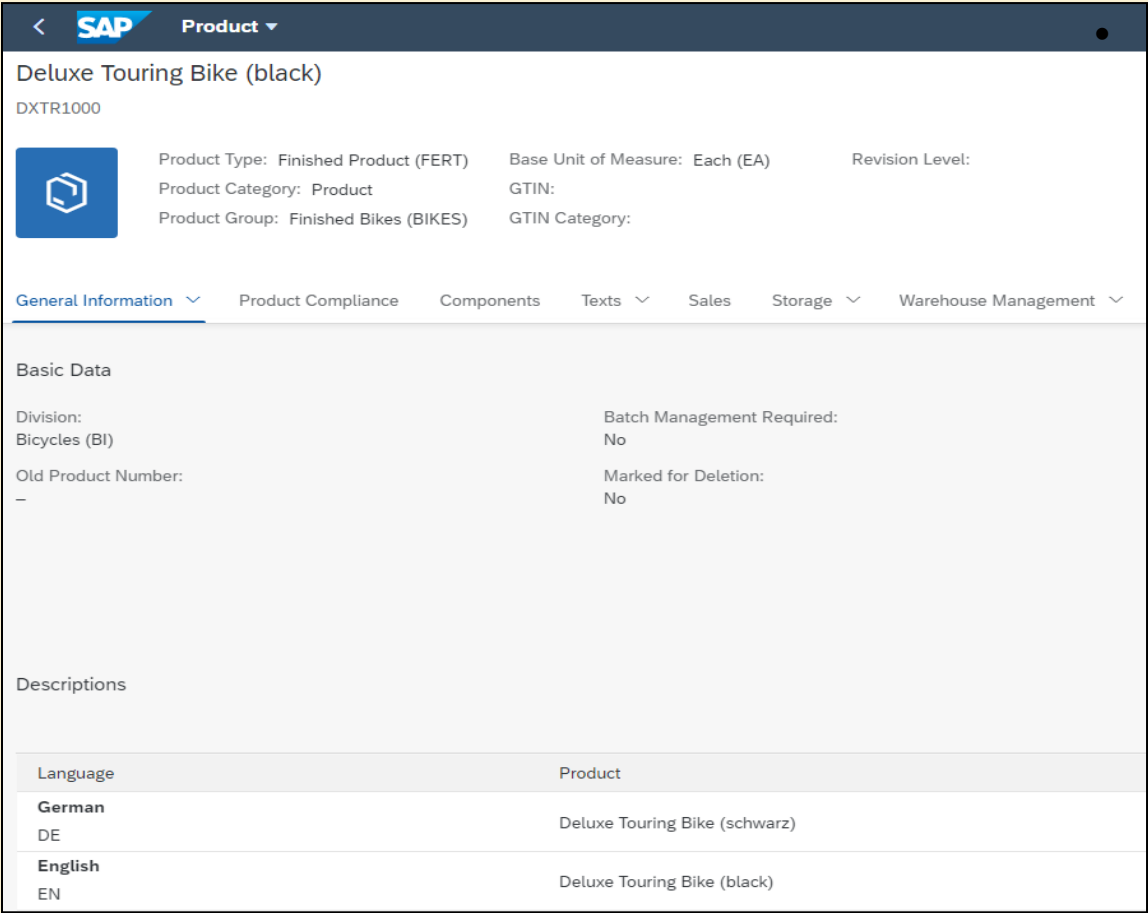


Sales Office Currency



Material Master Data

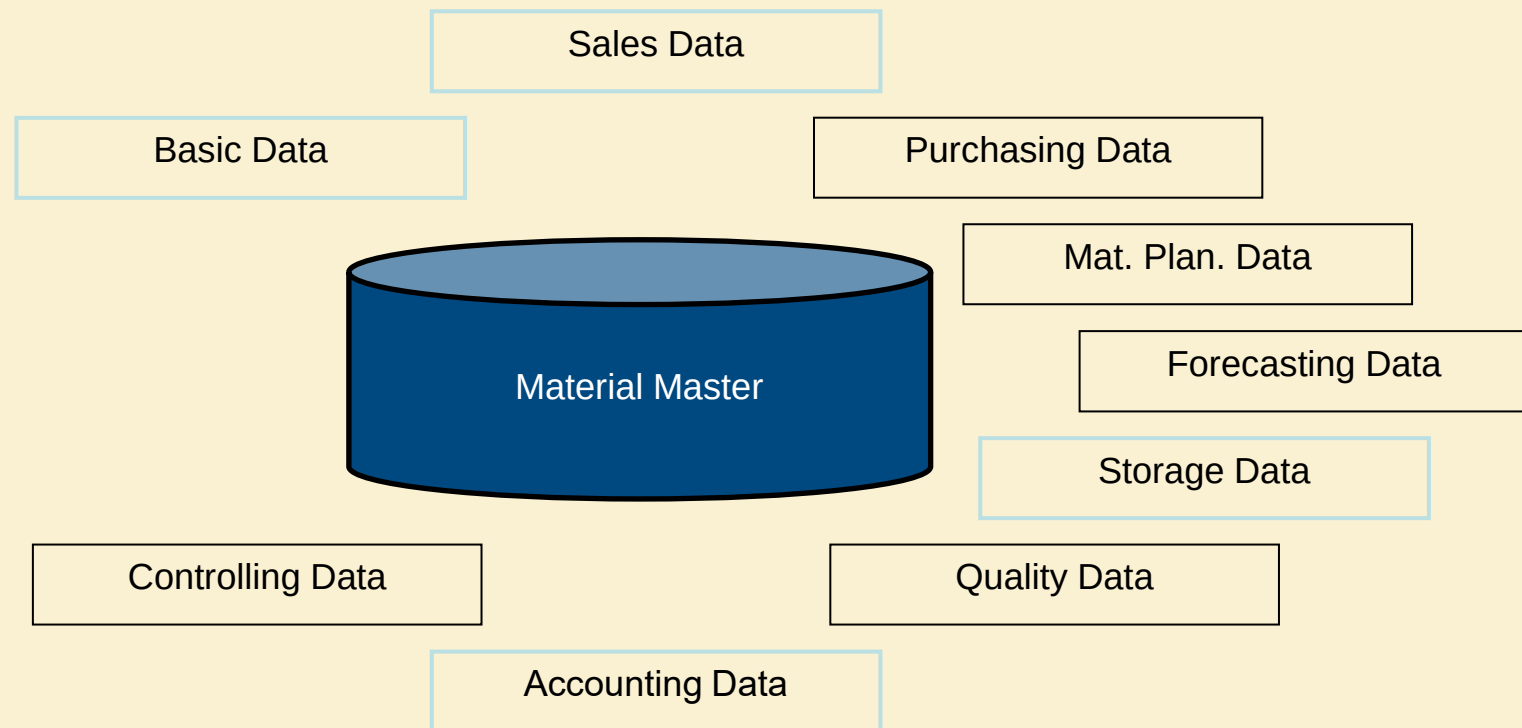
- Material Master contains all the information a company needs to manage about a material. Material master data is stored in functional segments called Views.



It is used by most components within the SAP system.

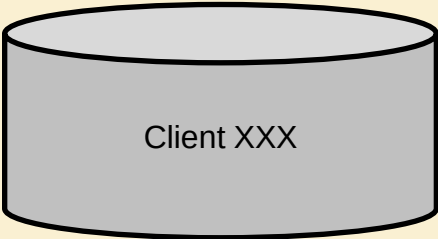
- ❖ Sales and Distribution.
- ❖ Materials Management.
- ❖ Production.
- ❖ Plant Maintenance.
- ❖ Accounting/Controlling.
- ❖ Quality Management.

Material Master Views



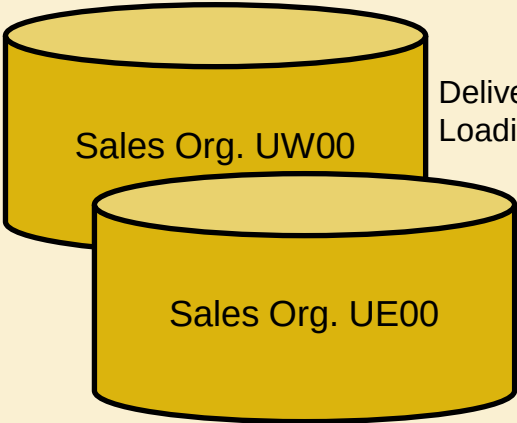
Material Master

General Information relevant for the entire organization:



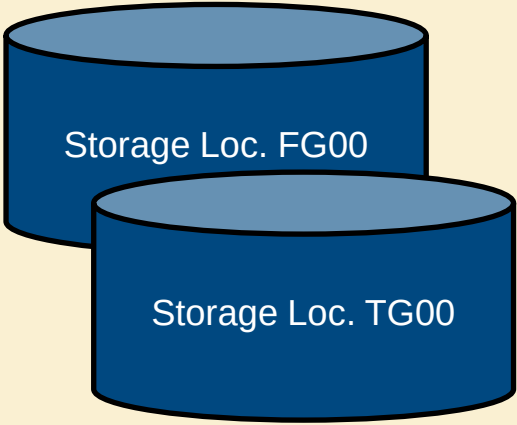
Name
Weight
Unit of Measure

Sales specific information:



Delivering Plant
Loading Grp

Storage Location specific information:



Stock Qty

Condition Master Data (Pricing)

• Condition master data includes:

- ❖ Prices.
- ❖ Surcharges.
- ❖ Discounts.
- ❖ Freights.
- ❖ Taxes.

<<<>>>

Sales Document Item: 10

Item category: TAN

Standard Item

Material: DXTR1000

Deluxe Touring Bike (black)

Sales A

Sales B

Shipping

Billing Document

Conditions

Account Assignment

Schedule lines

Partner

Texts

Order

Quantity: 5 EA

Net: 14,012.50 USD

Tax: 0.00

63 Condition Record

Analysis

Update

	I...	CnTy	Description	Amount	Crcy	per	UoM	Condition Value	Curr.	Status	NumC...
☐	☒	PR00	Price	3,000.00	USD	1	EA	15,000.00	USD		1
☐			Gross Value	3,000.00	USD	1	EA	15,000.00	USD		1
☐	☒	K004	Material	50.00-	USD	1	EA	250.00-	USD		1
☐	☒	RA00	% Discount from Net	5.000-	%			737.50-	USD		0
☐			Discount Amount	197.50-	USD	1	EA	987.50-	USD		1
☐			Rebate Basis	2,802.50	USD	1	EA	14,012.50	USD		1
☐			Net Value for Item	2,802.50	USD	1	EA	14,012.50	USD		1
☐			Net Value 2	2,802.50	USD	1	EA	14,012.50	USD		1
☐			Total	2,802.50	USD	1	EA	14,012.50	USD		1
☐	☒	SKTO	Cash Discount	0.000	%			0.00	USD		0
☐	☒	VPRS	Internal price	1,400.00	USD	1	EA	7,000.00	USD		1
☐			Standard - USA /With	1,402.50	USD	1	EA	7,012.50	USD		1

- You can define the condition master to be dependent on various data:
 - ❖ Material specific.
 - ❖ Customer specific.
- Conditions can be dependent on any document field.

Output

- Output is information that is sent to the customer using various media, such as:
 - ❖ Electronic Data Interchange (EDI).
 - ❖ Email.
 - ❖ Mail.
 - ❖ Fax.
 - ❖ XML.
- Functions for sales, shipping, transportation, and billing.

Output (Cont.)

- Examples of output types in SD processing are order confirmations, freight lists, and invoices
 - ❖ The output type defines how the output should be transmitted (sent via EDI, or as a print-out).
- Generation of
 - ❖ Sales activity output (e.g. customer telephone calls, mailing campaigns).
 - ❖ Group output (e.g. freight lists).

In conclusion

- Sales and Service.
- Master Data and its role in ERP Systems.
- Creating Customer Master Data, Material Master Data, and Pricing Conditions.

Understand



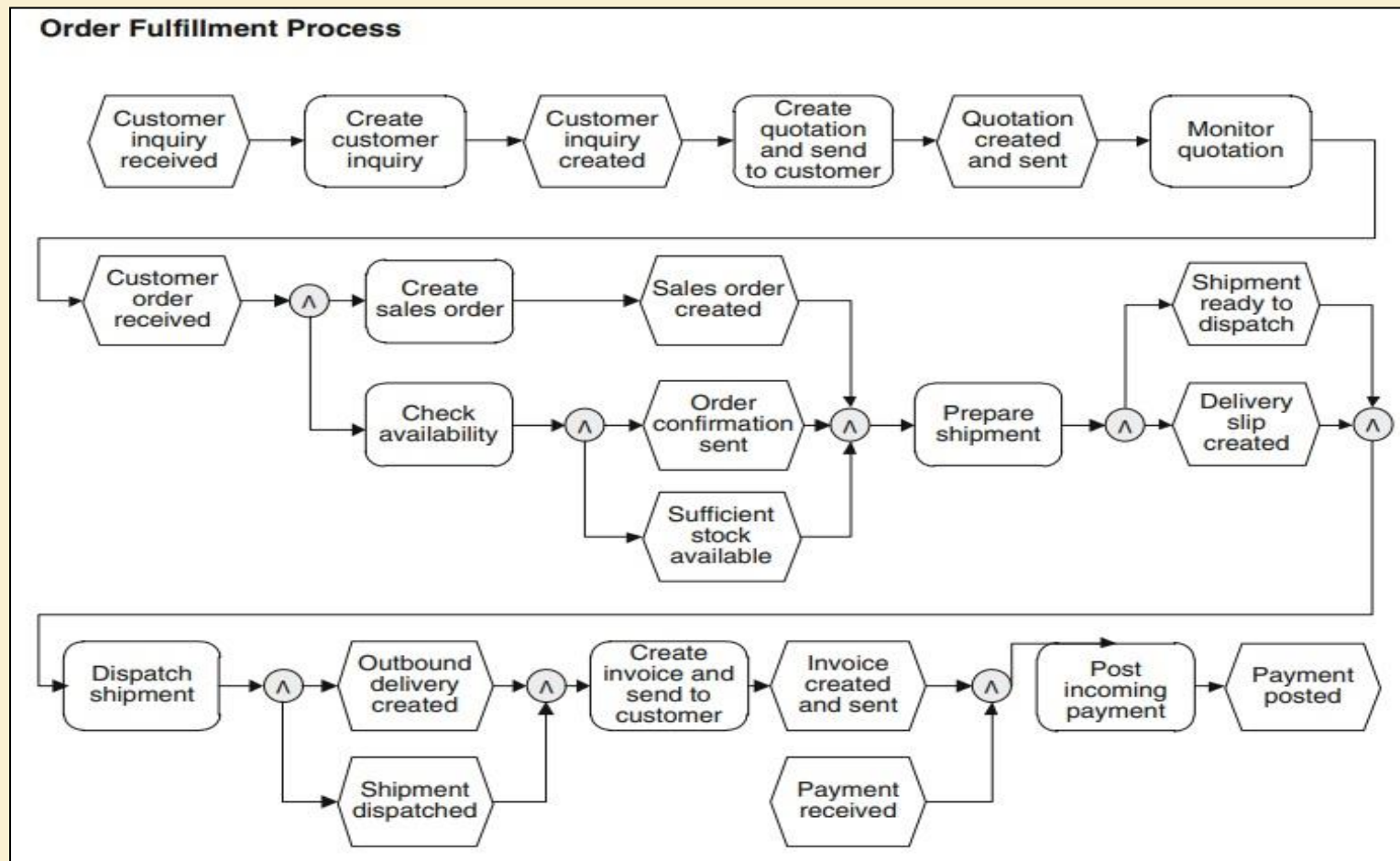
4. Sales and Fulfillment Cycle

What is Order Fulfillment Process?

- The term “order fulfillment” refers to the process of filling customer orders, starting with the first inquiry, all the way to the shipment of the goods, and receiving the customer’s payment.
- For this process, functions of sales, material management, and accounting are needed.
- In most cases, the process does not start with a customer order but with an inquiry in which the customer asks for certain goods (or services), prices, delivery dates, terms, and conditions.

What is Order Fulfillment Process? (Cont.)

- The order fulfillment process



Sales & Distribution Organizational Structure

- Client
 - ❖ An independent environment in the system.
- Company Code
 - ❖ Smallest organizational unit for which you can maintain a legal set of books.
- Credit Control Area
 - ❖ An organizational entity which grants and monitors a credit limit for customers.
 - ❖ It can include one or more company codes.

Sales & Distribution

Organizational Structure (Cont.)

- Sales Organization
 - ❖ An organizational unit responsible for the sale of certain products or services.
 - ❖ The responsibility of a sales organization may include legal liability for products and customer claims.
- Distribution Channel
 - ❖ The way in which products or services reach the customer.
 - ❖ Typical examples of distribution channels are wholesale, retail, or direct sales.

Sales & Distribution Organizational Structure (Cont.)

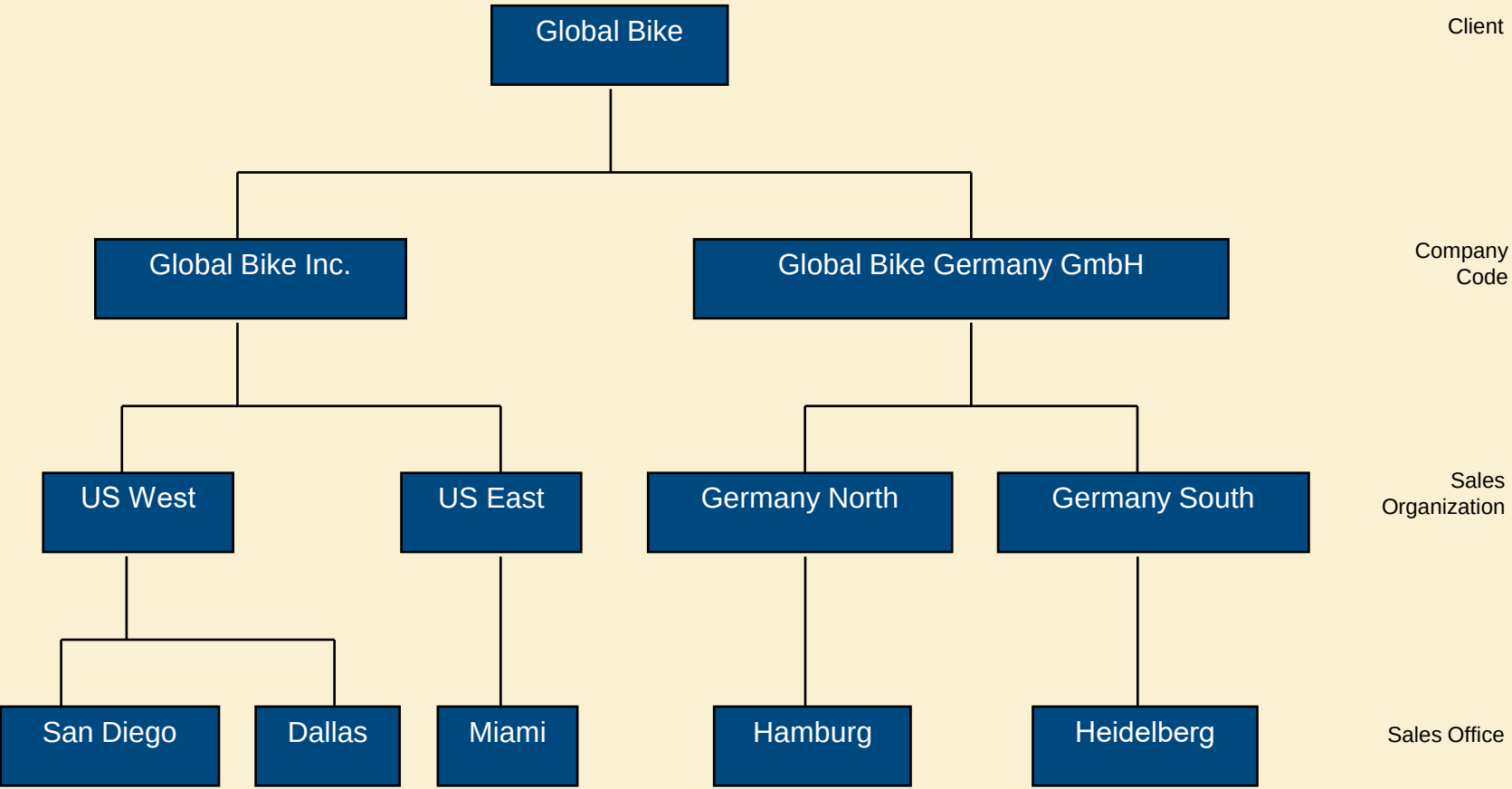
- Division
 - ❖ A way of grouping materials, products, or services.
- Sales Area
 - ❖ Combination of Sales Organization, Distribution Channel and Division.
 - ❖ Determines conditions (i.e. pricing) for sales activities.

Sales & Distribution

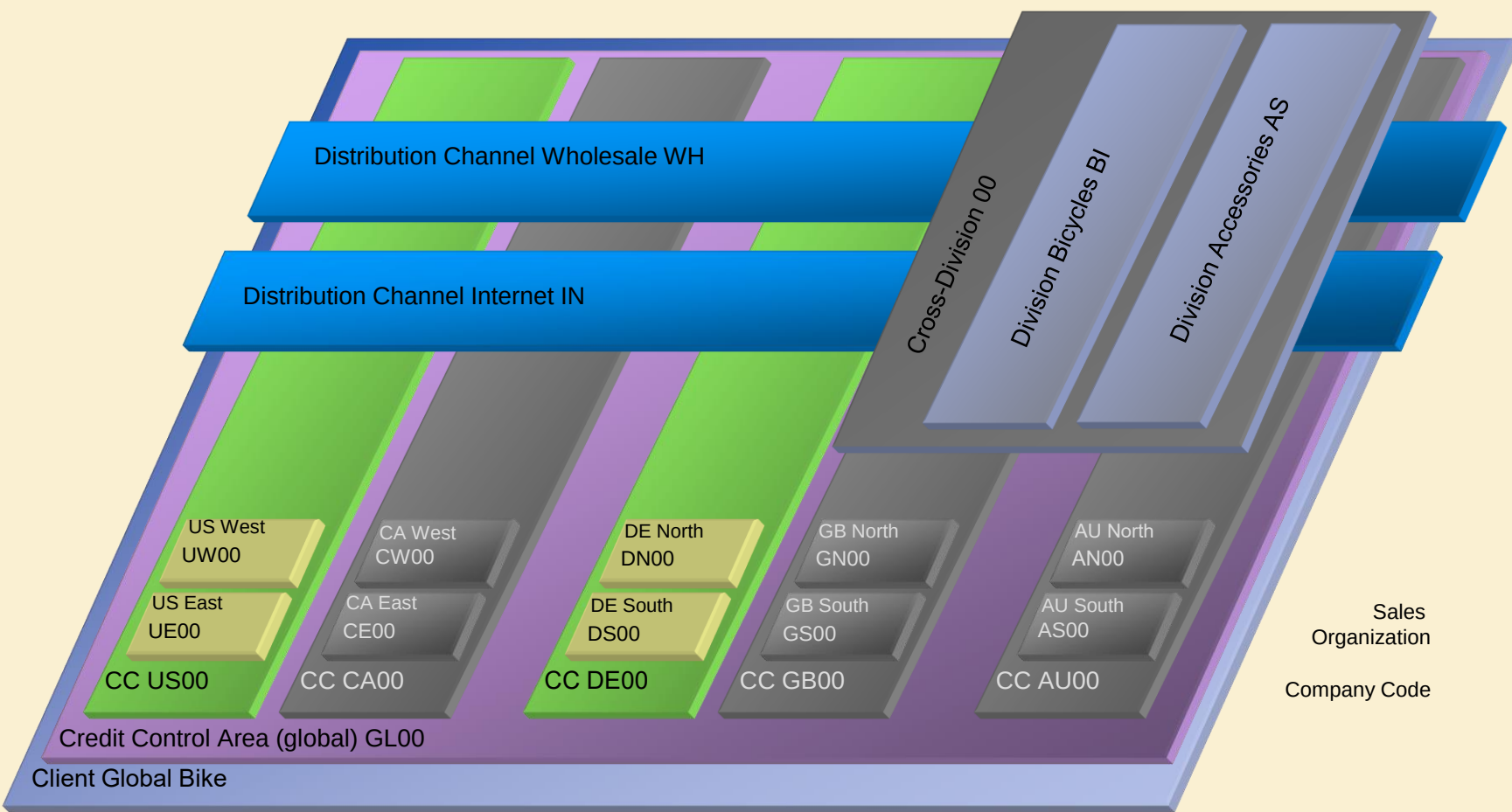
Organizational Structure (Cont.)

- Plant (here: Delivering Plant)
 - ❖ Plant from which the goods should be delivered to the customer
- Other SD organization units:
 - ❖ Shipping Point.
 - ❖ Loading Point.
 - ❖ Sales Office.
 - ❖ Sales Group.
 - ❖ Sales Person.

Global Bike Structure for Sales and Distribution



Global Bike Enterprise Structure in SAP ERP (Sales)



Sales Order Process

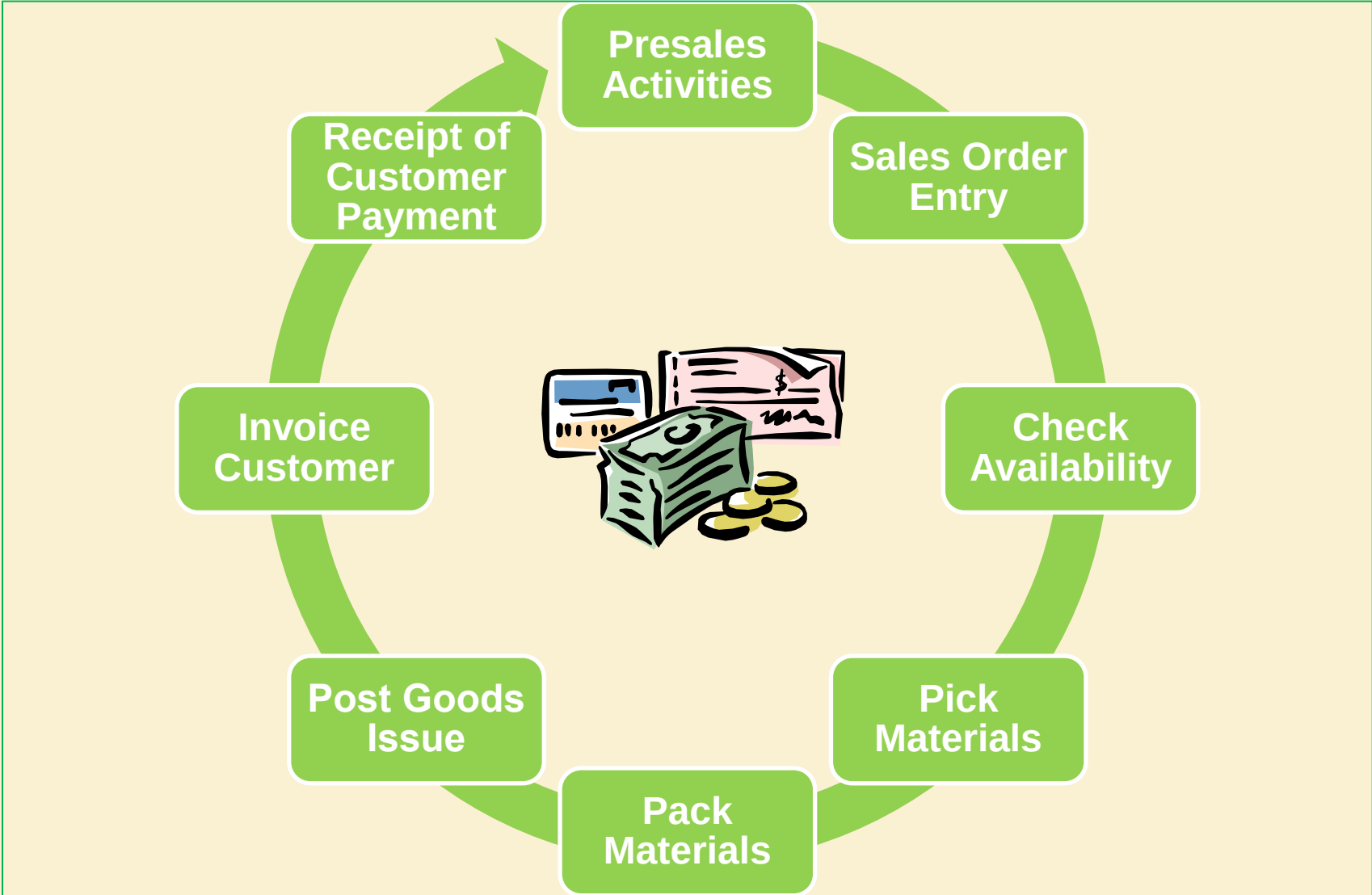


Figure 1: Sales Order Process. Source: Introduction to SAP S4HANA @2023 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Pre-Sales Activities (CRM Light)

- **Sales Support** is a component of SD that assists the processes of sales, distribution, and marketing of companies as well as their product and services to its customers.
- It contains the following functionality
 - ❖ Creating and tracking customer contacts and communications (sales activity).
 - ❖ Implementing and tracking direct mailing, internet, and trade fair campaigns based on customer attributes.

Pre-Sales Activities (CRM Light) (Cont.)

- Pre-sales documents: ***Inquiries and Quotations.***
- The ultimate goal of all presales activities is to equip the sales technician with all the **information** necessary to negotiate and complete the potential sale.
 - ❖ Past sales activity.
 - ❖ Past communication.
 - ❖ Contact information.
 - ❖ General company info.
 - ❖ Credit limits and usage.
 - ❖ Current backorders.

Pre-sales documents - Inquiry

- An inquiry is a customer’s request to a company for information or quotation concerning their products or services without any obligation to purchase
 - ❖ How much will it cost?
 - ❖ Material/Service availability.
 - ❖ It may contain specific quantities and dates.
- The system maintains the inquiry, creating a quotation to address questions for the potential customer.

Company
The Bike Zone
2105 N Lewis Ave
Portland OR 97227

Inquiry

Number/Date
10000002 / 04/21/2010
Reference no./Date
000 /04/21/2010
Delivery date
Day 04/21/2010
Cust. no.
1301
Validity period
04/21/2010 bis 05/21/2010

We deliver according to the following conditions:
Terms of payment Payable immediately without deduction

Currency USD

Terms of delivery FOB San Diego

Weights (gross/net) - Volume - Mark
Gross weight 57,170 G Net weight 57,170 G

Please see our promotional offer enclosed. Delivery as long as stocks last.

Item	Material	Qty	Description	Price	Price unit	Value
000010	DXTR1000	5 EA	Deluxe Touring Bike (black)	3,000.00	USD	15,000.00
000020	PRTR1000	2 EA	Professional Touring Bike (black)	3,200.00	USD	6,400.00
Final amount						21,400.00

Pre-sales documents - Quotation

- The quotation presents the customer with a legally binding offer
 - ❖ to deliver specific products or a selection of a certain amount of products.
 - ❖ in a specified timeframe.
 - ❖ at a predefined price.
- Legally binding on the vendor
- Conditions can apply at various levels:
 - ❖ to the entire quotation
 - ❖ at item level, to the material to be supplied or to the planned procurement in the case of services.
 - ❖ at service-line level for individual services (tasks or activities).

Company
The Bike Zone
2105 N Lewis Ave
Portland OR 97227

Quotation
Number/Date
20000000 / 04/21/2010
Reference no./Date
000 /04/21/2010
Delivery date
Day 04/21/2010
Cust. no.
1301
Validity period
04/21/2010 bis 05/21/2010

We deliver according to the following conditions:
Terms of payment Payable immediately without deduction

Currency USD

Terms of delivery

FOB San Diego

Weights (gross/net) - Volume - Mark

Gross weight 57,170 G Net weight 57,170 G

Please see our promotional offer enclosed. Delivery as long as stocks last.

Item	Material	Qty	Description	Price	Price unit	Value
000010	DXTR1000		Deluxe Touring Bike (black)			
		5 EA	3,000.00	USD	1 EA	15,000.00
	Material		50.00-	USD	1 EA	250.00-
	% Discount from N		5.000-	%		737.50-
	Net Value for It		2,802.50	USD	1 EA	14,012.50
000020	PRTR1000		Professional Touring Bike (black)			
		2 EA	3,200.00	USD	1 EA	6,400.00
	% Discount from N		5.000-	%		320.00-
	Net Value for It		3,040.00	USD	1 EA	6,080.00
Final amount						20,092.50

Sales Order

- Sales order processing can originate from a variety of documents and activities
 - Customer contacts us for an order: phone, internet, e-mail
 - Existing Contract
 - Quotations
- The electronic document that is created should contain the following basic information:
 - Customer Information
 - Material/service and quantity
 - Pricing (conditions)
 - Specific delivery dates and quantities
 - Shipping information
 - Billing Information

Sales Order (Cont.)

- The sales document is made up of three primary areas:

- ❖ Header

- Data relevant for the entire sales order: e.g.: customer data, total cost of the order.

- ❖ Line Item(s)

- Information about the specific product: e.g.: material and quantity, cost of an individual line.

- ❖ Schedule Line(s)

- Uniquely belongs to a Line Item, contains delivery quantities and dates for partial deliveries.

Display Standard Order 5: Overview

Standard Order: 5

Net Value: 20,092.50 USD

Sold-To Party: 1003063

The Bike Zone 000 / 2144 N Orange Ave / Orlando FL 32804

Ship-To Party: 1003063

The Bike Zone 000 / 2144 N Orange Ave / Orlando FL 32804

Cust. Reference: 000

Cust. Ref. Date: 09/14/2021

Sales

Item Overview

Item detail

Ordering party

Procurement

Shipping

Reason for rejection

Req. Deliv.Date: D 10/14/2021

Deliver.Plant:

Complete Dlv: ☐

Total Weight: 57,170 G

Delivery Block:

Volume: 0.000

Billing Block:

Pricing Date: 09/14/2021

Pyt Terms: 0001 Pay immediately w/o deduction

Inco. Version:

Incoterms: FOB

Inco. Location1: Miami

All Items

Item	Material	Req. Segment	Order Quantity	Un	S	Item Description
☐	10 DXTR1000		5	EA	☐	Deluxe Touring Bike (black)
☐	20 PRTR1000		2	EA	☐	Professional Touring Bike (black)

Sales A

Sales B

Shipping

Billing Document

Conditions

Account Assignment

Schedule lines

Partner

Texts

Order Data

Status

Structure

Fixed Date and Qty: ☐

Order Quantity: 5 EA

Delivery Time:

Delivered qty: 5

Quantities/Dates

P..	Delivery Date	Order Quantity	Rounded qty	Confirmed Qty	Sa...	Delivery Block	Delivered qty	Sch...	Purchase Re...	Requ...
☐	D 10/14/2021	5	5		5 EA			5 CP		0

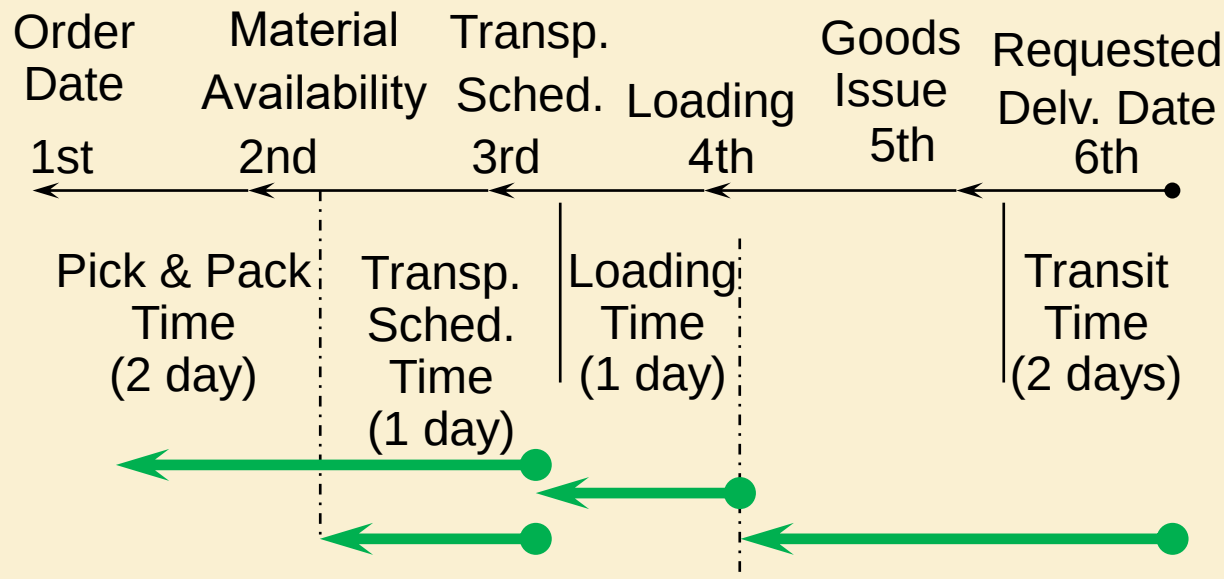
Sales Order (Cont.)

- The sales order contains all of the information needed to process your customers' request. The following information is determined for each sales order:
 - ❖ Delivering schedule.
 - ❖ Shipping point and route determination.
 - ❖ Availability check.
 - ❖ Transfer of requirements to MRP.
 - ❖ Pricing.
 - ❖ Credit limit check.

Sales Order - Delivery Scheduling

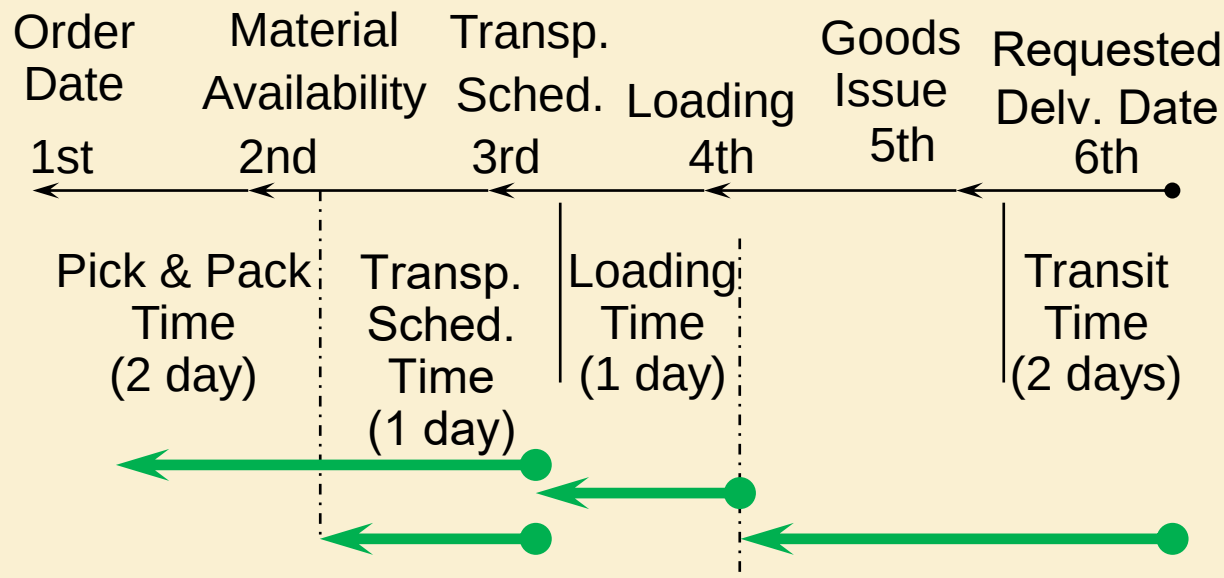
- When an order is created, you must enter a requested delivery date for the order or each line item.
- The system will then determine a delivery timeline, which will be used when determining our material availability, or ATP (Availability to Promise) date.
- The system will determine this date using the forward and backward scheduling rules you have defined.

Sales Order - Backward Scheduling



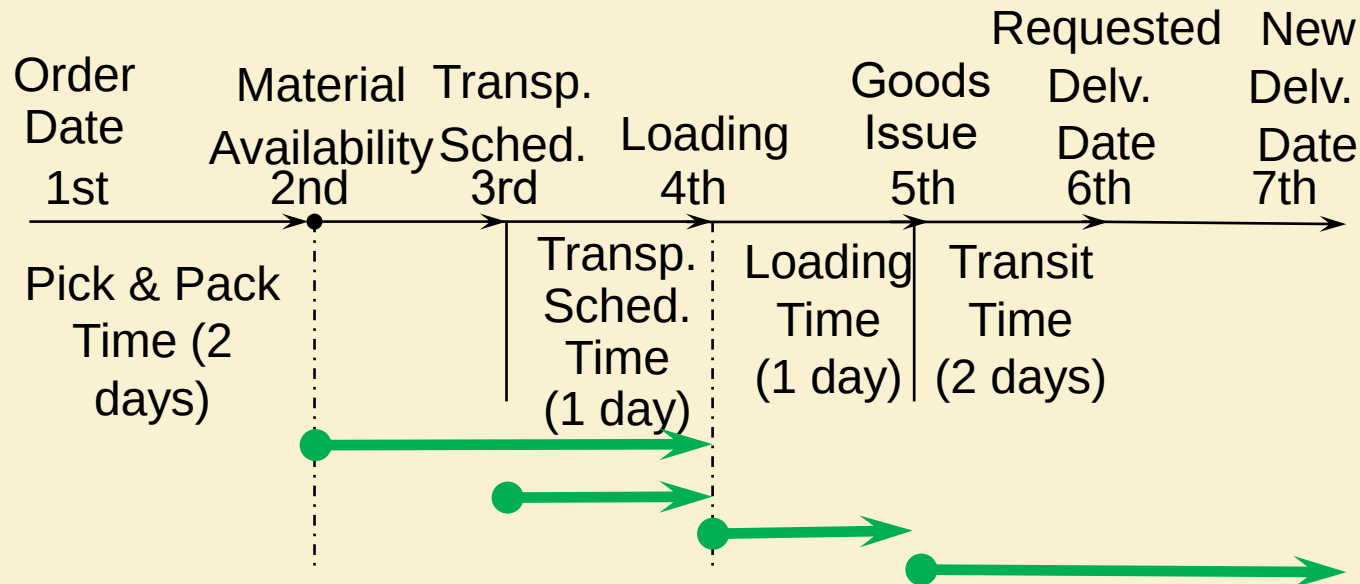
- Backward scheduling is the calculation of deadline dates: the arrival time at the customer site is calculated as the earliest possible goods receipt time at the customer's unloading point on the requested delivery date.
- All four of the delivery and transportation scheduling lead times are subtracted from the customer's requested delivery date to determine if this date can be met.

Sales Order - Backward Scheduling (Cont.)



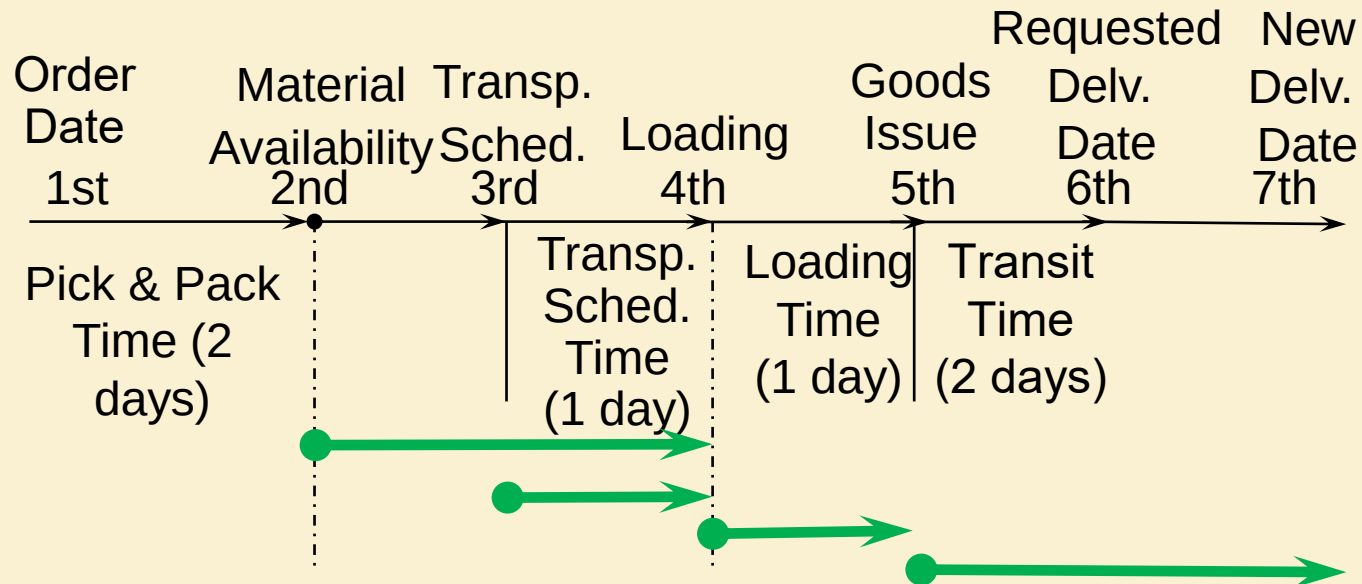
- The **transit time** is the time in days that is required to deliver goods from your premises to the customer location. It is defined for a route.
- The **loading time** is the time in days that is required for loading a delivery item. It is determined from the shipping point, the route, and the loading group of the material.
- The **pick/pack time** is the time in days that is required for allocating goods to a delivery as well as the time in days that is required for picking and packing. It is calculated using the shipping point, the route, and the weight group of the order item.
- The **transportation lead time** is the time in days that is needed to organize the shipping of the goods. This might include booking a ship and reserving a truck from a forwarding agent. It is defined for a route.

Sales Order - Forward Scheduling



- Forward scheduling is also done if no product is available on the material availability date calculated by backward scheduling. The system does an availability check to determine the first possible date when the product will be available.
- This new material availability date forms the starting point for scheduling the remaining activities. The loading time, pick/pack time, transit time, and transportation lead time are added to the new material availability date to calculate the confirmed delivery date.

Sales Order - Forward Scheduling (Cont.)



- The **transit time** is the time in days that is required to deliver goods from your premises to the customer location. It is defined for a route.
- The **loading time** is the time in days that is required for loading a delivery item. It is determined from the shipping point, the route, and the loading group of the material.
- The **pick/pack time** is the time in days that is required for allocating goods to a delivery as well as the time in days that is required for picking and packing. It is calculated using the shipping point, the route, and the weight group of the order item.
- The **transportation lead time** is the time in days that is needed to organize the shipping of the goods. This might include booking a ship and reserving a truck from a forwarding agent. It is defined for a route.

Sales Order - Shipping & Route Determination

- During the creation of the sales order, the system must determine the shipping point from which the material will be shipped and the route the material will take to get from your warehouse to your customer's location.
- A shipping point is determined for each line item within the order.
- The route determination is used to define the transit time of the material that we used in scheduling.

Sales Order - Availability Check

- Availability Check
 - ❖ Determines the material availability date.
 - ❖ Considers all inward and outward inventory movements.
- Proposes 3 methods of delivery
 - ❖ One-time delivery.
 - ❖ Complete delivery.
 - ❖ Delayed proposal.
- Rules are created by USERS.

< **SAP** Availability Overview

Σ

⌵

Stock Overview

Stock in Detail

Totals Overview

Totals in Detail

Scope of check

More

Material: PRTR1000

Professional Touring Bike (black)

Plant: MI00

MRP Area: MI00

Base Unit: EA

Avail. check: 02

Check.rule: A

With reqmts qtys: ☐

End lead time: 09/21/2021

Totals display

Receipts: 78

Issues: 0

Confrmd issues: 0

ATP situation

	Date	MRP e...	Stock Segment	MRP element data	Rec./reqd qty	Confirmed	Cum. ATP qty
☐	09/16/2021	Stock			78		78
☐	09/16/2021	SLocSt		FG00	78		78
☐							

Sales Order - Pricing

- The system displays pricing information for all sales documents on the pricing screens at both the header and the line item level.
 - ❖ Header pricing is valid for the whole order, and it is the cumulative of all line items within the order.
 - ❖ Line item pricing is for each specific material.
- The system will automatically search for prices, discounts, surcharges, calculated taxes, and freight.
- You can manually manipulate the pricing at both the header and line item levels within the sales order by entering a condition type and amount.
 - ❖ Taxes and freight can be set up automatically, so we cannot manually enter them.

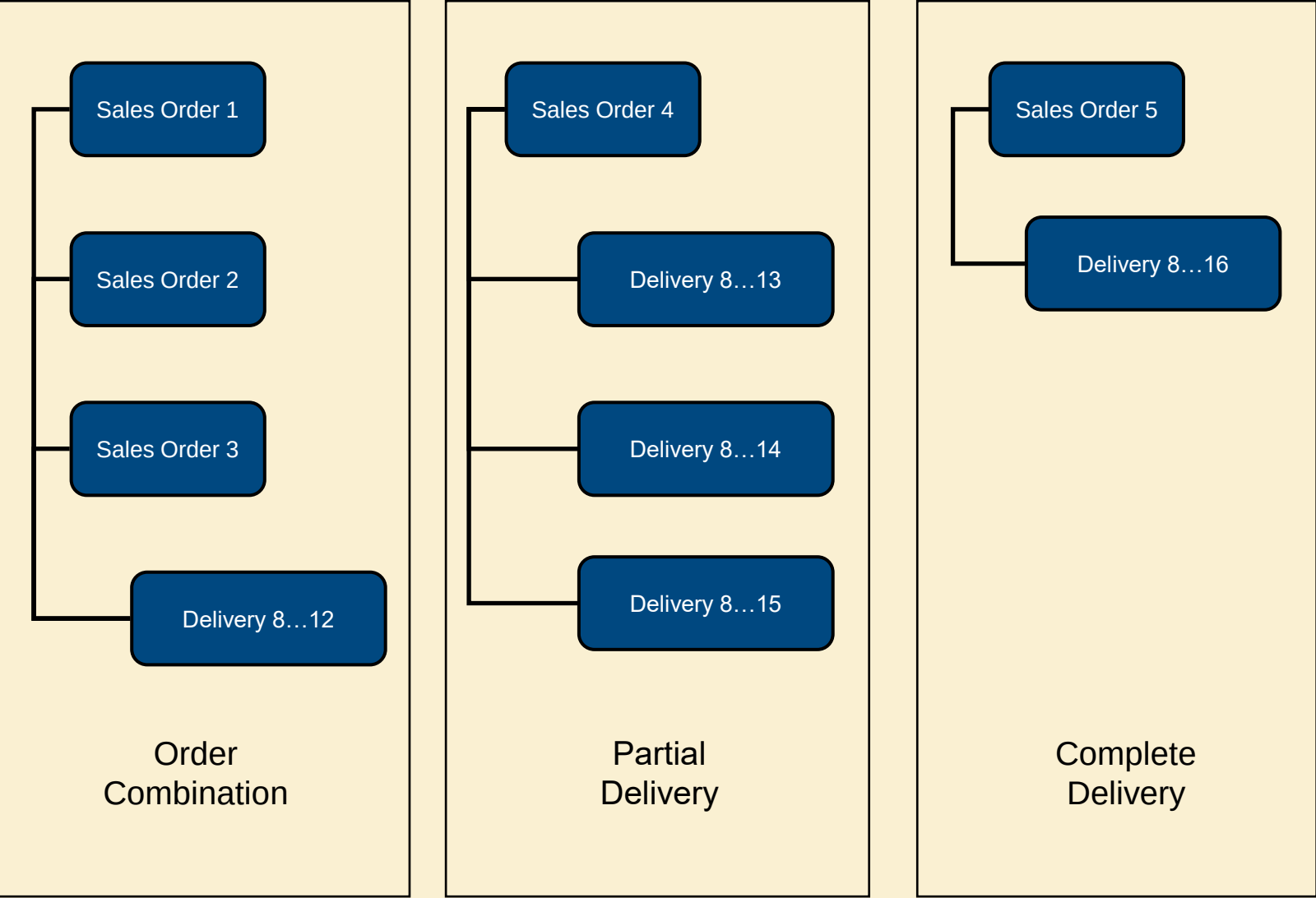
Sales Order - Credit Check

- Allows your company to manage its credit exposure and risk for each customer by specifying credit limits.
- During the sales order process, the system will alert the sales report about the customers' credit situation that arises.
- If necessary, the system can be configured to block orders and deliveries.

Shipping & Transportation

- The shipping process begins when you create the delivery document for the sales order. This document controls, supports, and monitors numerous sub-processes for progressing the shipping:
 - ❖ Picking.
 - ❖ Packing.
 - ❖ Post Goods Issue.
- Integrated with the Material Management (MM), extended Warehouse Management (eWM) and Finance (FI) modules.

Shipping



Delivery Creation

- Checks order and materials to determine if a delivery is possible — delivery block (hold), completeness.
- Confirms availability.
- Determines total weight & volume.
- Generates packing proposal.
- Checks scheduling.
- Considers partial deliveries.
- Updates route assignment.
- Assigns picking location.
- Updates sales order.
- Determines batches.
- Quality check (if needed).
- Updates sales order.

Delivery Document

- The Delivery Document initiates the delivery process and is the control mechanism for the following processes:
 - ❖ Picking.
 - ❖ Packing.
 - ❖ Loading.
 - ❖ Posting Goods Issue.
- Changes to delivery are allowable - products, quantities.

Picking

- Quantities based on delivery note.
- Assigned date when picking should begin.
- Automated storage location assignment.
- Supports serial number/lot number tracking and batch management.
- Integrated with Warehouse Management (WM).

Loading and Packing

- Identifies which packaging is to be used for specified products (customer preference and UCC-128 considerations).
- Identifies and updates accounts associated with returnable packaging.
- Tracks the packed product by container.
- Ensures weight/volume restrictions are enforced.
- All packed items are assigned to the required means of transportation.

Goods issue

- Event that could indicates the legal change in ownership of the products.
 - ❖ It signals that the goods have left the premises of the seller. Legal change in ownership depends on the “Incoterm” and what is written in the sales order terms and conditions.
- Reduces inventory and enters Cost of Goods Sold.
- Automatically updates the General Ledger (G/L) accounts.
- Ends the shipping process and updates the status of the shipping documents.

Billing

- The billing document is created by copying data from the sales order and/or delivery document.
 - ❖ Order-based billing.
 - ❖ Delivery-based billing.
- The billing process is used to generate the *customer invoice*.
- It will update the customer's credit status.
- The *billing document* will automatically create a debit posting to your customer's sub-ledger account and credit it to your revenue account.
- It is at this point that the sales process is passed over to Financial Accounting to await payment.

Billing Methods

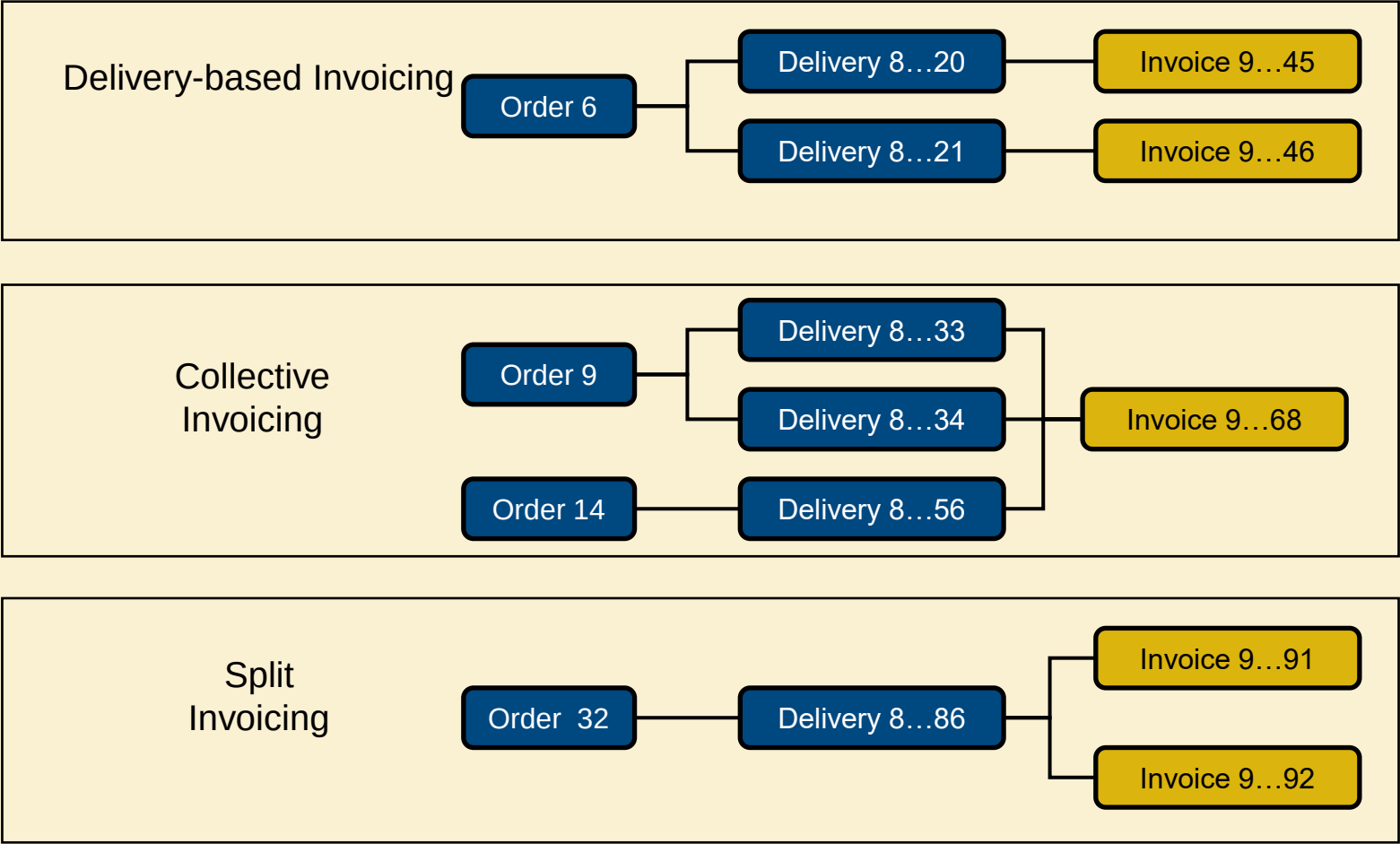


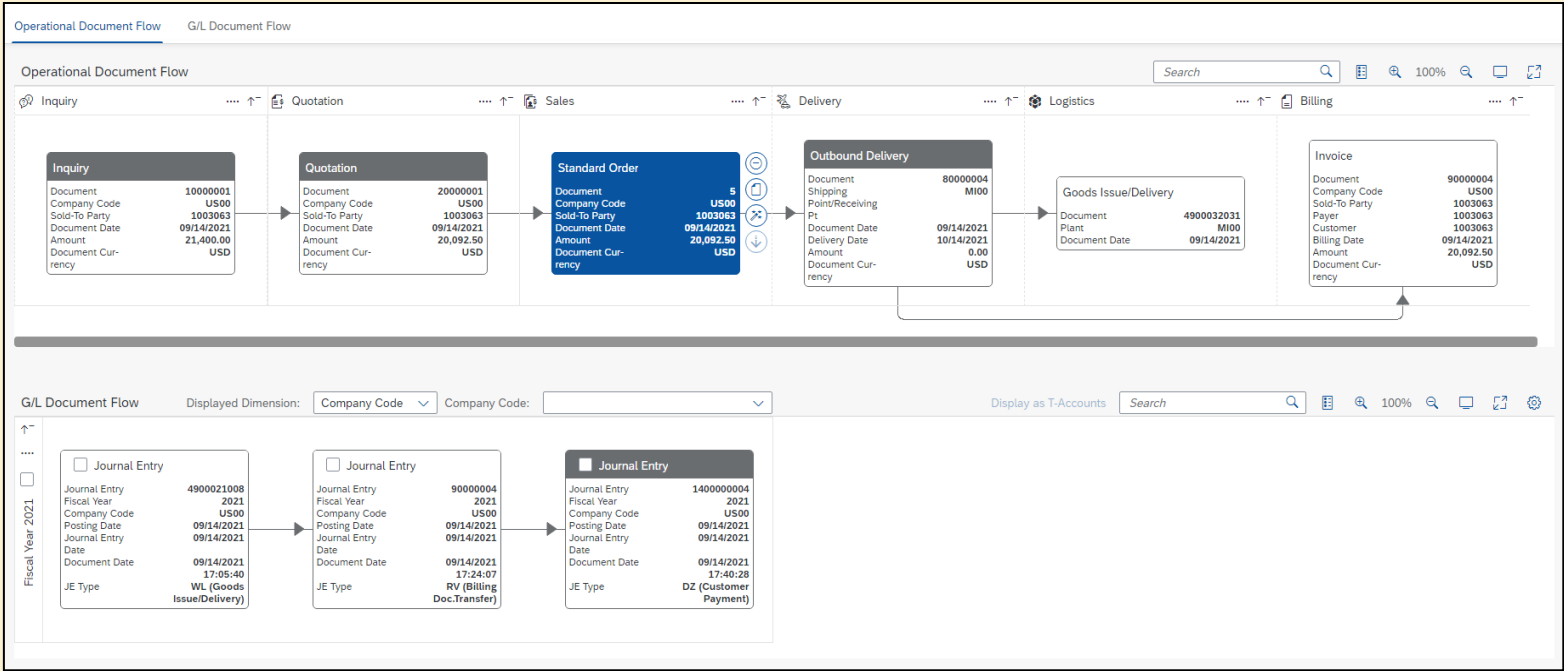
Figure 3: Billing Methods. Source: Introduction to SAP S4HANA @2022 SAP SE/SAP UCC Magdeburg. All rights Reserved.

Payment

- Payment is the final step in the sales order process, this step is managed by the Financial Accounting department.
- Final payment includes
 - ❖ Posting payments against invoices.
 - ❖ Reconciling differences between payment and invoice.
- Payment will create a posting clearing the liability in the A/R account and increases your bank account.

Document Flow

- The document flow and order status feature allows you to find the status of an order at any point in time. The SAP updates the order status every time a change is made to any document created in the customer order management cycle (Order-to-Cash).

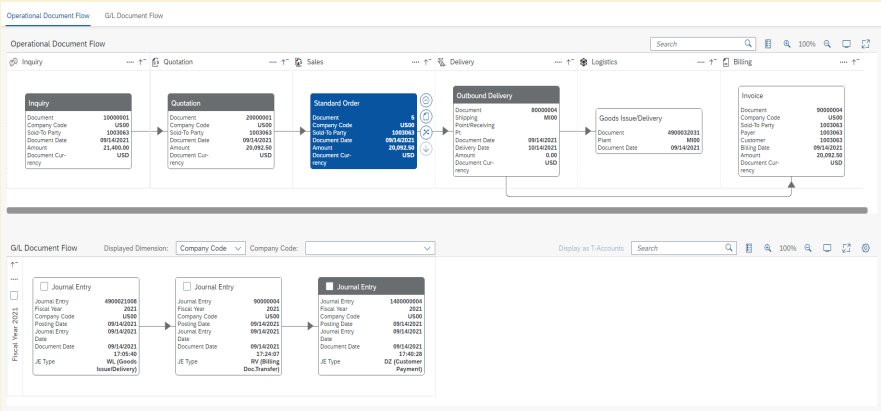


Sales Order Process Debugging

- “Never again did I learn so much in such a short time, because twenty participants made mistakes for me! You cannot make so many mistakes all alone”

SAP Co-founder Klaus Tschira
on his experiences teaching COBOL
to clients at IBM

- Document Flow 
 - ❖ Gives Order Process Status
- List of Sales Orders (VA05)
 - ❖ Tool to Find Order
 - ❖ Need student’s user id/data set number



In conclusion

- Sales Order Process
- Pre-Sales Activities (CRM Light)
- Sales Order
- Shipping & Transportation
- Billing
- Payment
- ...

Understand



5. Sales and Distribution in SAP S/4HANA (using Global Bike)

**THANK YOU
FOR YOUR ATTENTION**

Q&A